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EVALUATING THE FIDELITY AND COMPLETENESS OF MULTI-STYLE LLM-GENERATED EXPLANATIONS FOR AI- BASED PHISHING DETECTION

by

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ABSTRACT

One of the most persistent cybersecurity threats that individuals and organizations globally face is phishing, and while AI-based detection systems have showcased outstanding classification accuracy, the reasoning behind their decision is hardly communicated in a way that end users understand. This gap is addressed in this study by evaluating the fidelity and quality of multi-style LLM-generated explanations for AI based phishing detection.

A Random Forest classifier was trained on a phishing email dataset that was publicly available, obtaining an accuracy of 99.48% and a recall of 99.45%. SHAP values were retrieved from the trained model to serve as its reasoning, and three differently styled explanations, namely technical, simple and narrative, were generated using the Claude API for a sample of 300 emails, generating 900 explanations in total. These were automatically evaluated, using a multi-metric framework evaluating fidelity, readability, accuracy, clarity and completeness, with statistical significance examined using the Kruskal-Wallis test.

The findings identified statistically significant differences between all three explanation styles across all five evaluation metrics. The technical style obtained the strongest fidelity score of 0.788, demonstrating its alignment with the model's underlying reasoning, but performed badly across all other metrics. The narrative style generated the most readable outputs but underperformed on accuracy and completeness. The simple explanation style was identified as the most effective overall, obtaining the highest scores for accuracy, clarity and completeness while keeping a positive readability score.

These findings contribute greatly to the Explainable AI and cybersecurity literature by presenting the first systematic, automated comparison of multi-style LLM generated explanation in a phishing detection setting. The study gives a recommendation that developers of AI-driven phishing detection systems should use plain language explanations as the default communication style for general user audiences, while providing adaptive explanation options for more technical users.

Keywords: Phishing Detection, Explainable AI, Large Language Models, SHAP, Explanation Evaluation, Natural Language Generation, Cybersecurity.

CHAPTER 1: INTRODUCTION

This chapter gives a thorough summary of the research study. It begins by establishing the aim of the study, followed by the background section that places in context the increasing problem of phishing detection and the limitations of existing AI-based systems with respect to explainability. This is followed by the specific research questions that govern the investigation. Next, the objectives of the study are defined, outlining the specific steps taken to resolve the research questions. The chapter ends by providing a description of the overall structure of the report, giving the reader a defined plan of what ensues in the following chapters.

1.1 AIMS

The overall aim of this study is to evaluate the fidelity and quality of multi-style LLM generated explanations for AI-based phishing detection. This is accomplished by building an AI-based phishing classifier, producing three separate explanation styles (technical, simple and narrative) using a Large Language Model (LLM) and carefully evaluating each style through an automated multi-metric assessment model.

The study aims to tackle a known gap in the Explainable AI (XAI) literature, where explanation generation has attained significant attention but the comparative evaluation of explanation styles is still largely unexplored, especially in the cybersecurity field. By focusing on fidelity, completeness, readability and overall quality as evaluation criteria, the study is looking to produce findings that are both academically thorough and practically significant to the design of transparent AI-driven phishing detection systems.

1.2 BACKGROUND

Threats in Cybersecurity continue to evolve at a rapid rate, with phishing maintaining its position as one of the most relentless and damaging attack vectors that individuals and organizations face worldwide. Phishing often defined as a trick-based approach used to deceive users into revealing sensitive data by impersonating trusted parties, has increased significantly in size and sophistication. The Anti-Phishing Working Group (APWG) states that the third quarter of 2022 by itself recorded over 1.27 million phishing attacks globally, the highest quarterly total ever recorded at the time. As the size of these attacks increases, the need for intelligent, automated detection systems that are able to identify threats faster than human intervention alone also increases.

To address this problem, machine learning (ML) and deep learning (DL) solutions have emerged as the dominant framework for phishing detection. Work carried out by Basit et al. (2021) and Salloum et al. (2022) provides thorough evaluations of AI-enabled phishing detection techniques, showcasing that ML based classifiers, especially transformer-based models such as BERT, continuously outperform traditional rule-based and heuristic methods in both accuracy and adaptability. Although these advances exist, a critical limitation remains.

The majority of AI-based phishing detection systems function like black boxes. They produce a classification output, either phishing or legitimate, but provide minimal to no explanation of reasoning behind that decision. This ambiguity is a well-known problem in the broader field of Artificial Intelligence, and it is exactly this problem that the field of Explainable Ai (XAI) intends to address.

XAI has grown at a rapid rate into a key area of research, fuelled by the demand to make AI decision-making transparent, readable and trustworthy. Arrieta et al. (2020) provide a primary classification of XAI concepts, opportunities and challenges, establishing that explainability is not just a feature to be desired but a necessary condition for responsible AI deployment, especially in high-stakes environments such as cybersecurity. Tools such as SHAP (Shapley Additive exPlanations) and LIME (Local Interpretable Model Agnostic Explanations) have been used to extract feature-level explanations from trained models, as established by Kamath and Liu (2021) and Molnar (2022). In recent times, the emergence of Large Language Models (LLMs) has pioneered a new frontier in explanations generation, the ability to transform raw model reasoning into natural language descriptions that can be communicated to a broad audience.

A significant gap exists in the literature despite this progress. While research has examined how to produce AI explanations, minimal work has studied how the style of those explanations their fidelity and quality in showcasing the model's actual reasoning. Islam et al. (2022) showcase that explanation evaluation remains one of the least explored domains in XAI research, with majority of studies placing focus on explanation generation rather than the structured evaluation of explanation quality. Within this particular context of phishing detection, the gap is even more significant. No study to date has systematically compared multiple LLM generated explanation styles against a thorough automated evaluation framework.

This study directly addresses that gap, by building an AI-based phishing detection classifier and generating three distinct explanation styles ; technical, simple and narrative, using an LLM. This research proposes and adopts a novel multi-metric automated evaluation framework assessing each style across four criteria: fidelity to model reasoning, completeness of phishing indicator coverage, readability and LLM as a judge scoring. As a result, this study supports both the XAI and cybersecurity literature by providing the first systematic, automated comparison of multi-style LLM-generated explanations in the context of phishing detection. The results aim to provide evidence-based guidance to researchers and developers on how AI phishing detection should communicate their reasoning, eventually giving support to the development of more transparent and trustworthy cybersecurity tools.

1.3 RESEARCH QUESTIONS

This study follows the guidance of the following research questions:

RQ1: To what extent do LLM-generated explanations across three distinct styles, namely technical, simple, and narrative, differ in their fidelity to the underlying reasoning of the phishing detection classifier?

RQ2: How do the three explanation styles perform under LLM-as-Judge evaluation in terms of accuracy, clarity, and overall quality?

RQ3: Which LLM-generated explanation style is most effective overall in communicating the reasoning of an AI-based phishing detection system, based on the combined findings of the automated evaluation framework?

These research questions collectively influence the study's direction, proceeding logically from the review of current literature through to the development of the classifier, the generation of explanations and their comparative evaluation. Each question aligns directly to the objectives outlined in the following section.

1.4 OBJECTIVES

The following objectives have been established to help provide guidance for the study and to address the research questions that have been stated above:

O1: To critically review current literature on phishing detection, Explainable AI (XAI) and LLM-generated explanation evaluation in order to identify the existing research gaps and build a theoretical foundation for the study.

O2: To collect, preprocess, and prepare a suitable publicly available phishing email dataset for training and evaluating a machine learning classifier.

O3: To develop and evaluate a Random Forest classification model capable of detecting phishing emails with measurable and reportable performance metrics.

O4: To extract model reasoning from the trained classifier using SHAP values and generate three stylistically distinct LLM-produced explanations, namely technical, simple, and narrative, for a representative sample of 300 emails.

O5: To automatically evaluate the fidelity, readability, accuracy, clarity, and completeness of each explanation style using a multi-metric automated evaluation framework.

O6: To statistically compare the three explanation styles across all evaluation metrics and determine which style most effectively communicates the reasoning of an AI-based phishing detection system.

1.5 STRUCTURE OF THE REPORT

This report is organised across six chapters and a reference list, each fulfilling a specific and purposeful role in addressing the research questions and meeting the overall aim of the study.

Chapter One: Introduction establishes the foundation of the research by presenting the background to the study, articulating the research aim, formulating the research questions and objectives, and outlining the overall structure of the report.

Chapter Two: Literature Review provides a critical and systematic examination of existing scholarly work across three interconnected domains: phishing detection methodologies, Explainable Artificial Intelligence (XAI), and Large Language Model-based explanation generation. The chapter synthesises key findings from the literature and situates the present study within the existing body of knowledge, clearly identifying the gap this research seeks to address.

Chapter Three: Research Methodology and Design of Practical Work presents a detailed account of the research design and procedural framework adopted in the study. It covers the rationale for dataset selection, data pre-processing procedures, classifier development and evaluation, explanation generation using an LLM, and the automated multi-metric evaluation framework employed to assess explanation fidelity and quality.

Chapter Four: Data Analysis, Findings and Implementation and Testing presents the empirical findings of the study in a structured and systematic manner. It reports the classification performance of the trained model, the evaluation scores obtained across all three explanation styles, and documents the implementation and testing process. Statistical analysis is applied to determine the significance of observed differences, and findings are supported through appropriate data visualisations.

Chapter Five: Evaluation and Implications critically evaluates the outcomes of the study in relation to the research questions and the broader literature. It reflects on the effectiveness of the automated evaluation framework, discusses the practical implications of the findings for the design of transparent AI-driven cybersecurity systems, and acknowledges the limitations encountered throughout the research process.

Chapter Six: Conclusions, Recommendations and Self-Reflection provides a concise summary of the study's key findings and contributions to the fields of Explainable AI and cybersecurity. It offers evidence-based recommendations for future research and concludes with a personal reflection on the research process, the challenges encountered, and the skills developed throughout the study.

Reference List contains a comprehensive list of all sources cited throughout the report, formatted in accordance with the required academic referencing style.

CHAPTER 2: LITERATURE REVIEW

2.1 INTRODUCTION

This chapter presents a critical and systematic review of existing literature across three interconnected domains that collectively form the theoretical and empirical foundation of this study: phishing detection using artificial intelligence, Explainable Artificial Intelligence (XAI) and its evaluation frameworks, and the application of Large Language Models (LLMs) in generating natural language explanations. The review draws upon a range of peer reviewed journal articles, books, and conference proceedings published predominantly from 2020 onwards, reflecting the most current and relevant state of research in these rapidly evolving fields. Where foundational works predate this period, they are included on the basis of their sustained influence and theoretical significance. The chapter is structured thematically, moving from the broader context of phishing as a cybersecurity threat, through to the technical approaches adopted for its detection, before examining the principles and limitations of explainability in AI and the emerging role of LLMs in communicating model reasoning. The chapter concludes by synthesising these themes to identify the specific gap in the literature that this study directly seeks to address.

2.2 PHISHING AS A CYBERSECURITY THREAT

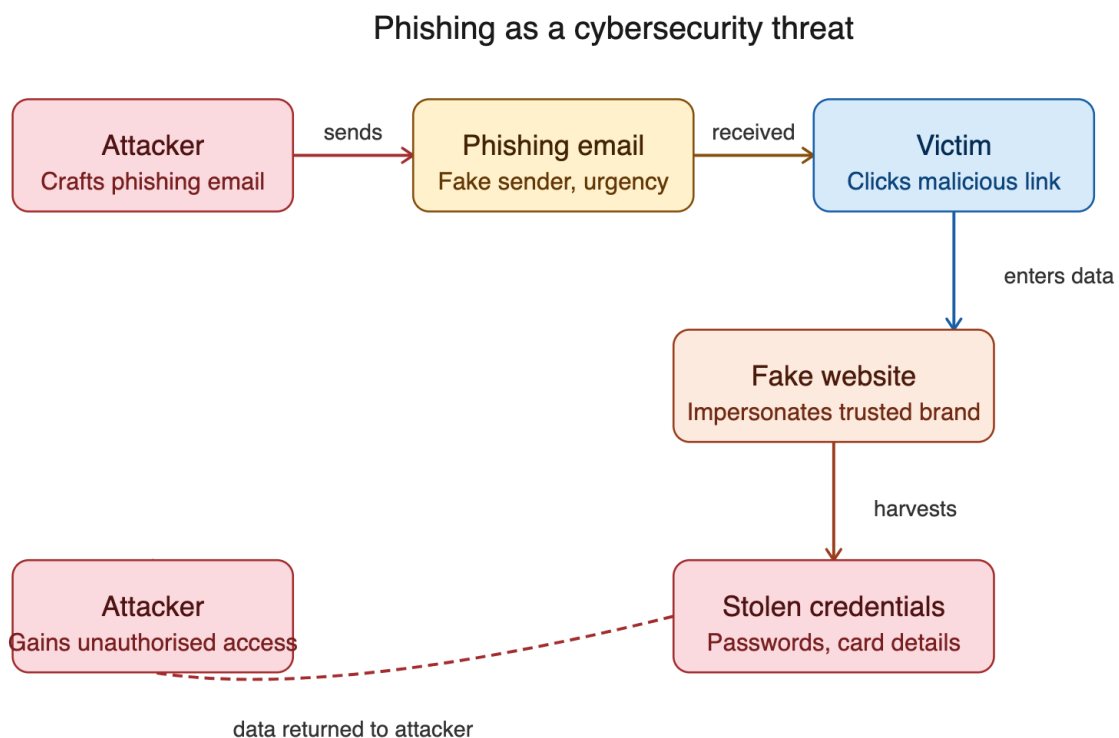


Fig2. 1: Phishing as a Cybersecurity Threat

Phishing has long been recognised as one of the most persistent and damaging forms of cybercrime, yet its continued evolution has made it considerably more difficult to combat using conventional security measures alone. At its core, phishing involves the use of deceptive

communication, most commonly delivered via email, to manipulate recipients into disclosing sensitive information such as login credentials or financial details by impersonating a trusted entity (Basit et al., 2021). What makes phishing particularly challenging from a detection standpoint is not simply its frequency but its adaptability. Attackers continuously refine their tactics to bypass existing defences, leveraging psychological manipulation, brand impersonation, and increasingly convincing language to exploit human trust at scale.

The scale of the problem has grown considerably in recent years. The Anti-Phishing Working Group (APWG) reported that the third quarter of 2022 alone recorded over 1.27 million phishing attacks globally, representing the highest quarterly total ever documented at the time. Financial institutions remained the most targeted sector, accounting for approximately 23% of all attacks during that period, while webmail and software as a service providers accounted for a further 17% (APWG, 2022). These figures point to a threat landscape that is not only persistent but accelerating, placing considerable pressure on researchers and developers to produce detection systems that are both accurate and scalable.

It is worth noting that phishing is not solely a technical problem but also a deeply human one. Shahriar, Mukherjee and Gnawali (2022) argue that phishing emails are deliberately engineered to exploit cognitive biases and psychological vulnerabilities, including authority bias, urgency, and fear, making them effective even against users with a reasonable degree of digital literacy. This observation has important implications for the design of detection systems. A classifier that flags a phishing email but provides no explanation of why it was flagged does little to address the underlying human vulnerability. Users who do not understand why a message is suspicious are unlikely to learn from the experience or become more vigilant over time, suggesting that explainability is not merely a technical concern but a user education imperative.

Traditional approaches to phishing detection, such as blacklisting, rule based filtering, and heuristic analysis, have proven increasingly inadequate in keeping pace with these developments. Blacklists require continuous manual updates and offer no protection against newly registered phishing domains, while rule based systems are easily circumvented by attackers who understand the detection criteria. It is against this backdrop that machine learning and deep learning have emerged as the dominant paradigm for automated phishing detection, as evidenced by the breadth of literature reviewed in the following section.

2.3 MACHINE LEARNING APPROACHES TO PHISHING DETECTION

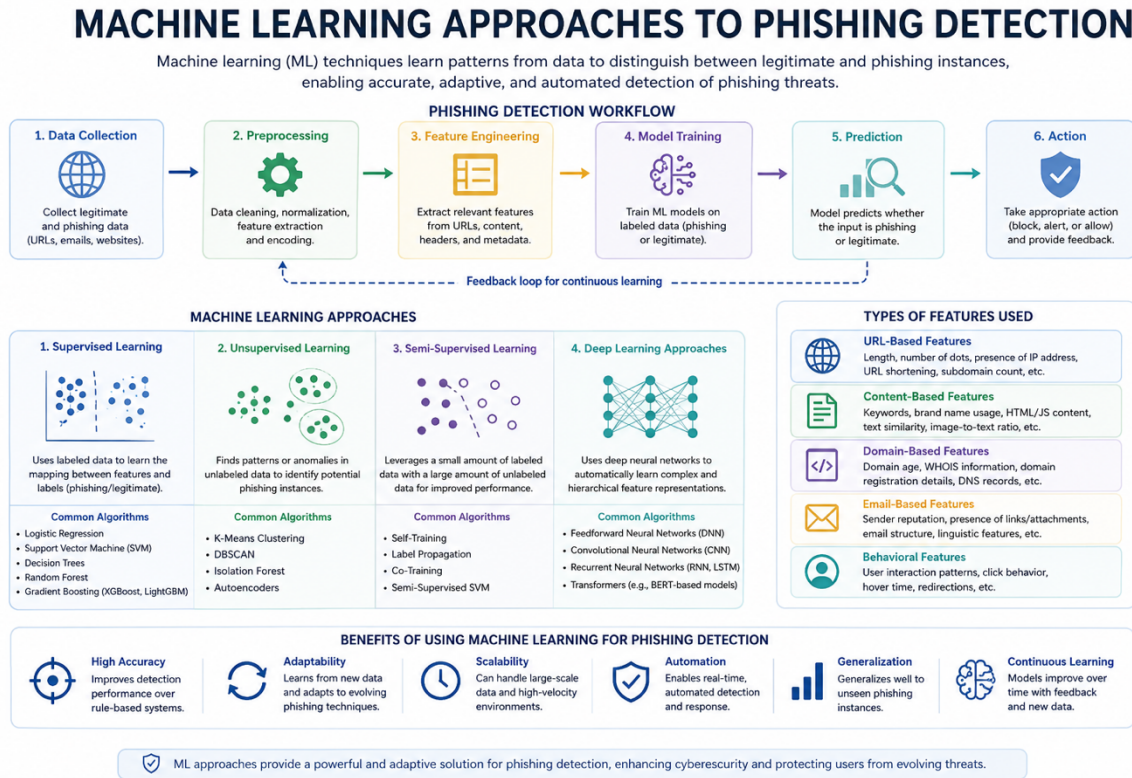


Fig2. 2: Machine Learning Approaches to Phishing Detection

The application of machine learning to phishing detection has been extensively documented in the literature, with a broad range of algorithms being evaluated across different detection contexts including URL analysis, email content classification, and web page visual similarity detection. Basit et al. (2021) conducted a comprehensive survey of AI enabled phishing detection techniques, covering approaches that target URLs, emails, and web pages. Their review found that ML based classifiers consistently outperformed traditional approaches in adaptability and generalisation, particularly when trained on sufficiently large and diverse datasets. The authors noted that no single algorithm dominates across all detection contexts, and that the choice of model should be informed by the specific characteristics of the data and the deployment environment.

Among the most widely studied algorithms in this domain are Random Forest, Support Vector Machines, Logistic Regression, and more recently, deep learning architectures including Convolutional Neural Networks, Long Short Term Memory networks, and transformer based models. Tang and Mahmoud (2021) conducted a survey of ML based solutions for phishing website detection, finding that ensemble methods such as Random Forest consistently demonstrated strong performance due to their robustness against overfitting and their ability to handle high dimensional feature spaces. However, the authors acknowledged that feature engineering, which involves the manual selection and extraction of relevant characteristics from URLs or email headers, remains a time consuming and domain dependent process that limits the generalisability of feature based approaches.

Salloum et al. (2022) carried out a systematic literature review of phishing email detection techniques using natural language processing, identifying BERT as one of the most effective models for email based classification due to its ability to capture contextual relationships within text that simpler bag of words or TF IDF approaches fail to represent adequately. This finding is consistent with broader research on transformer architectures, where Tunstall, von Werra and Wolf (2022) demonstrate that fine tuned BERT models consistently achieve high accuracy on text classification tasks across a range of domains, largely because pre training on large corpora allows the model to develop rich linguistic representations before being adapted to a specific task.

More recent comparative studies have reinforced the superiority of transformer based models over traditional ML classifiers in phishing detection specifically. Kapan and Sora Gunal (2023) reported that BERT and RoBERTa achieved detection accuracies of 98.99% and 99.08% respectively on a balanced merged dataset, outperforming traditional ML approaches by an average margin of 4.7%. While these results are impressive, the authors acknowledge that accuracy metrics alone do not capture the full picture of a model's utility in real world deployment, particularly when end users require an understanding of why a given decision was made. This critical observation is central to the motivation of the present study.

A further concern raised in the literature relates to model generalisation across different phishing datasets. Several studies, including those by Adebawale, Lwin and Hossain (2023) and Almomani et al. (2022), have demonstrated that models trained on one dataset frequently underperform when applied to another, raising questions about the robustness and transferability of existing detection approaches. Adebawale, Lwin and Hossain (2023) specifically found that deep learning models, while achieving high accuracy on training and in distribution test data, were more susceptible to this generalisation problem than simpler ML classifiers, suggesting that the relationship between model complexity and practical reliability is not straightforward. This observation further underscores the importance of transparency in AI based phishing detection. If practitioners cannot easily understand how a model makes its decisions, they are poorly placed to identify when and why it may be failing.

Zhou (2021) notes more broadly that as machine learning models grow in complexity, the trade off between predictive power and interpretability becomes increasingly difficult to manage, underscoring the need for dedicated explanation frameworks that operate alongside high performing classifiers. This tension between performance and transparency is a recurring theme throughout the XAI literature and serves as the conceptual bridge between the phishing detection domain and the broader field of explainable AI reviewed in the following section.

2.4 EXPLAINABLE ARTIFICIAL INTELLIGENCE (XAI)

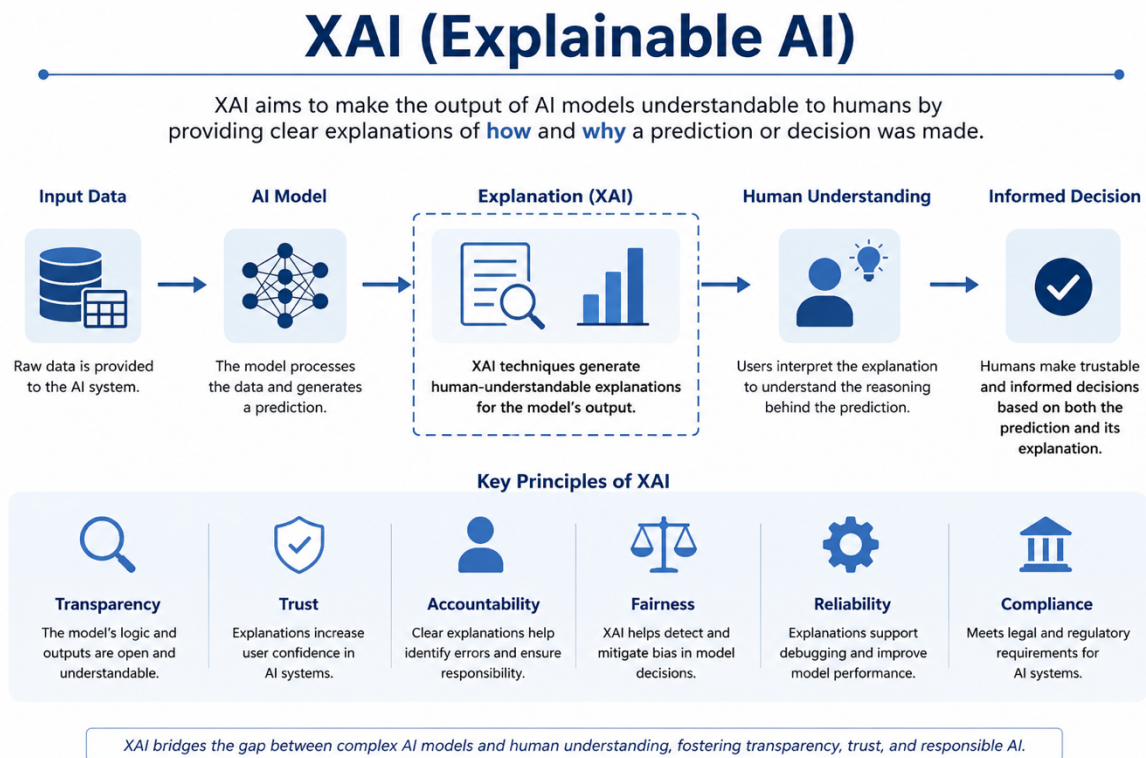


Fig2. 3: Explainable Artificial Intelligence

The concept of explainability in artificial intelligence has gained substantial traction over the past decade, driven by concerns over transparency, accountability, and trust in automated decision making systems. Arrieta et al. (2020) provide one of the most widely cited foundational frameworks in this space, offering a detailed taxonomy of XAI concepts, techniques, and challenges across multiple domains. Their work establishes that explainability is not merely a desirable feature of AI systems but a fundamental requirement for responsible deployment, particularly in high stakes domains such as healthcare, finance, and cybersecurity. The authors draw a useful and important distinction between interpretability, which refers to the degree to which a model's internal workings can be understood in terms of its architecture and parameters, and explainability, which refers to the ability to communicate those workings in terms that are meaningful and actionable for a human audience. This distinction is theoretically significant and practically consequential, as it implies that a model can be interpretable without being explainable in any useful sense.

Gunning et al. (2021) provide an informative retrospective on the DARPA Explainable AI programme, one of the most influential institutional efforts to advance XAI research. The programme identified three core goals for explainable AI systems: the ability to explain decisions to users, the ability to characterise the model's strengths and weaknesses, and the ability to convey an understanding of how the system will behave in the future. Notably, all three goals are fundamentally communicative in nature, reinforcing the view that explainability is as much about how information is conveyed as it is about what information is produced.

Two of the most widely adopted techniques for generating explanations from trained models are SHAP and LIME. Both approaches are model agnostic, meaning they can be applied to any trained classifier regardless of its internal architecture, which makes them practically versatile. SHAP assigns each input feature a contribution score based on principles derived from cooperative game theory, producing explanations that are both locally accurate and globally consistent (Molnar, 2022). LIME, by contrast, works by fitting a simpler interpretable model in the local neighbourhood of a given prediction, providing an approximation of the complex model's behaviour at that specific point. Kamath and Liu (2021) provide a thorough comparative analysis of these two approaches, noting that while SHAP offers stronger theoretical guarantees of consistency and accuracy, LIME tends to be more computationally efficient and easier to implement in practice, particularly for high dimensional data.

Despite the growing adoption of these tools, a number of significant limitations have been identified in the literature that are directly relevant to the present study. Rothman (2020) highlights that SHAP and LIME produce feature level explanations that, while technically informative to a data scientist, are often poorly suited to non technical audiences. A ranked list of feature importance scores or a set of highlighted text segments may satisfy the needs of a model developer but offers little practical guidance to an everyday user attempting to understand why an email was flagged as suspicious. This mismatch between the format of generated explanations and the informational needs of diverse end users has become an increasingly prominent concern in XAI research, and it represents one of the core motivations for exploring LLM based explanation generation as an alternative or complementary approach.

Hamm et al. (2023) investigated the effect of providing explanations on user trust and decision making in AI systems, finding that the presence of an explanation significantly influenced how users interacted with and responded to AI recommendations. However, the study also found that poorly constructed or misleading explanations could reduce trust and increase error rates, suggesting that not all explanations are equally beneficial and that the quality of an explanation matters as much as its presence. This finding has direct implications for the present study, as it provides empirical justification for systematically evaluating explanation quality rather than assuming that any explanation is inherently valuable.

Islam et al. (2022) conducted a systematic review of XAI across multiple application domains and tasks, finding that while explanation generation has received considerable research attention, the evaluation of explanation quality remains one of the least explored areas in the field. Most existing studies assess classifier performance in isolation, with minimal attention paid to whether the explanations produced are accurate, complete, or comprehensible to their intended audience. This gap is particularly evident in the cybersecurity domain, where the consequences of a poorly communicated explanation could leave a user more confused, or worse, less vigilant, than if no explanation had been provided at all.

2.5 EVALUATION OF AI GENERATED EXPLANATIONS

The evaluation of explanation quality is a complex and contested area of XAI research. Unlike classification performance, which can be measured through well established metrics such as accuracy, precision, recall, and F1 score, explanation quality is inherently more difficult to quantify objectively. Several frameworks and dimensions have been proposed in the literature, each approaching the problem from a different perspective and with different methodological assumptions.

Molnar (2022) identifies three broad dimensions along which explanations can be evaluated: fidelity, which captures how accurately the explanation reflects the model's true reasoning process; comprehensibility, which addresses how easily the explanation can be understood by its intended audience; and completeness, which concerns the extent to which the explanation accounts for all relevant factors that contributed to the model's decision. These dimensions provide a useful and widely referenced starting point for structuring an evaluation framework, though operationalising them in a rigorous and reproducible way remains an open challenge in the literature.

Hsieh et al. (2024) offer a more technically detailed treatment of explanation evaluation in their comprehensive guide to XAI, covering a range of automated evaluation metrics including faithfulness scores, sufficiency measures, and robustness tests across both classical models and LLMs. Their work is particularly relevant to the present study as it explicitly addresses the challenges of evaluating natural language explanations generated by LLMs, noting that the advent of natural language explanation generation introduces new evaluation demands that traditional XAI metrics, designed for feature importance scores and visual highlights, are not fully equipped to handle. This observation further strengthens the case for developing a novel evaluation framework tailored to the specific characteristics of LLM generated explanations.

Readability is a dimension of explanation quality that has received comparatively little attention in the XAI literature but is arguably of considerable practical importance, particularly in a cybersecurity context where explanations must communicate effectively with users who may have no technical background. The Flesch Kincaid readability formula, one of the most widely used and empirically validated measures of text readability, provides a quantitative score that reflects the average sentence length and word complexity of a given text (Rothman, 2020). While readability alone is insufficient as a measure of explanation quality, it provides a useful and reproducible dimension that complements more semantically oriented metrics such as fidelity and completeness.

One emerging approach to explanation evaluation that has gained traction in the NLP research community is the use of LLM as Judge evaluation, where a language model is prompted to assess the quality of a generated output against a defined rubric or set of criteria. While this method is relatively recent in its application to XAI evaluation, it has demonstrated considerable promise as a scalable and reproducible alternative to human evaluation, which is costly, time consuming, and subject to inter rater variability. Its specific application in the

context of evaluating explanation quality for phishing detection has not been explored in the existing literature, representing a clear and significant opportunity for research contribution.

2.6 LARGE LANGUAGE MODELS IN EXPLANATION GENERATION

The emergence of Large Language Models has opened new and significant possibilities for explanation generation in AI systems. Unlike SHAP and LIME, which produce structured numerical or visual outputs that require a degree of technical literacy to interpret, LLMs are capable of generating fluent, contextually appropriate natural language descriptions of model behaviour that can be tailored to different audiences and communication goals. This capability makes them particularly well suited to the task of communicating AI reasoning to audiences with varying levels of technical expertise, which is a challenge that traditional XAI tools have consistently struggled to address.

Tunstall, von Werra and Wolf (2022) provide a detailed account of how transformer based LLMs can be prompted to generate text across a range of styles, registers, and levels of complexity, highlighting the flexibility of these models as communication tools. In the context of XAI, this flexibility is valuable because it allows the same underlying model reasoning to be communicated in multiple ways, using technical terminology for a cybersecurity professional, plain language for a general user, or narrative framing for an audience with no prior knowledge of AI or phishing. The ability to generate stylistically distinct explanations from the same input represents a meaningful advance over traditional XAI approaches, though the question of which style is most effective in a given context has not yet been systematically addressed in the literature.

Research exploring the use of LLMs in cybersecurity contexts has expanded considerably in recent years, driven in large part by the availability of powerful general purpose models such as GPT-4 and Claude. Koide et al. (2024) demonstrated that LLMs can be prompted to both detect phishing emails and generate explanations of their classification decisions, achieving high precision and recall in controlled experiments. However, their study focused primarily on detection performance rather than the quality or comparative evaluation of the explanations generated, reflecting a broader tendency in the literature to prioritise classification metrics over communicative effectiveness.

Gupta et al. (2023) examined the broader implications of generative AI in cybersecurity, noting that while LLMs offer significant potential for enhancing detection and response capabilities, they also introduce new risks, including the possibility of generating explanations that are superficially plausible but factually inaccurate or misleading. This concern about hallucination in LLM generated outputs is directly relevant to the fidelity dimension of explanation evaluation, as it raises the possibility that a well written explanation may not accurately reflect the model's true reasoning. Systematically measuring this discrepancy, as this study proposes to do through the fidelity metric, is therefore not merely a technical exercise but a practically important safeguard against the deployment of explanations that misinform rather than illuminate.

The question of explanation style and its impact on communication effectiveness has received some attention in adjacent fields, including science communication, health literacy, and human computer interaction. Research in these areas has consistently found that the format, register, and framing of an explanation significantly affect how well it is understood, remembered, and acted upon by different audiences. Narrative framing, in particular, has been shown to enhance comprehension and engagement in contexts where the recipient lacks domain specific knowledge, as it provides a relatable context through which abstract information can be interpreted. Applying these insights to the specific context of AI generated phishing explanations represents a novel and underexplored research direction, and one that the present study directly addresses.

2.7 IDENTIFIED GAP IN THE LITERATURE

Table 2. 1: Identified Gap in the Literature

STUDY	CONTRIBUTION	IDENTIFIED GAP
Basit et al. (2021)	Extensive survey of AI enabled phishing detection methods across URLs, emails and web pages.	Does not tackle how model's reasoning is communicated to users.
Salloum et al. (2022)	Structured review of NLP methods for phishing email classification including BERT.	Focuses solely on classification performance and does not consider explainability.
Kapan and Sora Gunal (2023)	Comparative analysis of ML classifiers showcasing BERT and RoBERTa superiority.	Focuses solely on accuracy with no consideration for explanation generation or quality.
Arrieta et al. (2020)	Core classification framework of XAI concepts; opportunities and challenges.	Does not assess explanation styles or apply XAI evaluation to cybersecurity scenarios.
Islam et al. (2022)	Structured review of XAI across multiple fields specifically identifying explanation evaluation as under investigated.	Does not investigate phishing detection or compare LLM generated explanation styles.
Hamm et al. (2023)	Data-driven evidence that explanation quality influences user trust and decision-making results.	Does not apply findings to phishing detection or investigate whether explanation style affects quality outcomes.
Koide et al. (2024)	Showcases that LLMs can generate explanations for phishing classification decisions.	Explanations are not systematically assessed and there is no stylistic comparison.

Hsieh et al. (2024)	Showcases particular problems of evaluating natural language explanations generated by LLMs.	There is no evaluation framework for phishing detection and no comparison of explanation styles.
Gupta et al. (2023)	Thoroughly looks at the implications of generative AI in cybersecurity including hallucination risks.	Does not systematically assess or compare explanation quality across different styles.
Molnar (2022)	Systematic examination of fidelity, completeness and readability as explanation generation metrics.	Does not compare the explanation styles generated by LLMs or apply evaluation to phishing detection.

The review of existing literature reveals a clear, significant, and multi-dimensional gap at the intersection of XAI, LLM based explanation generation, and phishing detection. While considerable progress has been made in developing accurate AI based phishing classifiers, the explanations produced by these systems have received comparatively little scholarly scrutiny. The studies reviewed in this chapter demonstrate that explanation generation is technically feasible using both traditional XAI tools and LLMs, but none have systematically examined how different styles of LLM generated explanation compare in terms of fidelity, completeness, readability, and overall quality within a phishing detection context.

Islam et al. (2022) explicitly identify explanation evaluation as an underexplored area of XAI research, while Hsieh et al. (2024) draw attention to the particular challenges of evaluating natural language explanations generated by LLMs. Hamm et al. (2023) demonstrate empirically that explanation quality matters to end users and can influence both trust and decision making outcomes, yet no study to date has applied this insight to the specific domain of phishing detection or examined whether different explanation styles produce meaningfully different quality outcomes. Koide et al. (2024) show that LLMs can generate explanations for phishing classification decisions, but do not evaluate those explanations systematically or compare different stylistic approaches.

This study directly addresses the identified gap by proposing and applying a novel automated multi-metric evaluation framework that assesses three distinct LLM generated explanation styles, namely technical, simple, and narrative, across four evaluation dimensions: fidelity to model reasoning, completeness of phishing indicator coverage, readability as measured by the Flesch Kincaid formula, and overall quality as assessed through LLM as Judge scoring. In doing so, it makes a contribution that is both timely and practically relevant, offering evidence based guidance that could meaningfully inform the design of more transparent and trustworthy AI driven phishing detection systems.

2.8 SUMMARY

This chapter has critically reviewed the key literature underpinning this study, tracing the development of AI based phishing detection from traditional rule based approaches through to state of the art transformer models, examining the principles, tools, and limitations of Explainable AI, and exploring the emerging role of LLMs in natural language explanation generation. A consistent and significant gap has been identified across these three domains, namely the absence of any systematic, automated, and reproducible comparison of multi-style LLM generated explanations within a phishing detection context. The following chapter outlines the research methodology and practical design adopted to address this gap and answer the research questions set out in Chapter One.

CHAPTER 3: RESEARCH METHODOLOGY AND DESIGN OF PRACTICAL WORK

3.1 INTRODUCTION

This chapter outlines the research methodology and practical design adopted in this study. It covers the philosophical stance underpinning the research, the overall design rationale, and the procedures carried out across each phase of the investigation, including dataset selection and preprocessing, classifier development and evaluation, model reasoning extraction, explanation generation, and automated evaluation. Ethical considerations and measures taken to ensure the validity and reliability of the findings are also addressed.

3.2 RESEARCH PHILOSOPHY AND DESIGN

Every research study rests, whether explicitly acknowledged or not, on a set of philosophical assumptions about the nature of knowledge and the most appropriate means of acquiring it. Making these assumptions explicit is important because they shape every methodological decision that follows (Saunders, Lewis and Thornhill, 2023).

This study adopts a positivist philosophical stance. Positivism holds that knowledge is best produced through objective, systematic observation and measurement, and that findings should be replicable and generalisable beyond the immediate context of the investigation. This position is well suited to the present study, which is entirely computational and relies on quantitative metrics that are calculated automatically, applied consistently, and free from subjective interpretation. The use of numerically defined evaluation dimensions, including fidelity ratios, completeness ratios, Flesch Kincaid scores, and structured LLM as Judge rubrics, reflects a commitment to objectivity that is consistent with positivist epistemology (Bryman, 2021).

A quantitative research approach follows naturally from this philosophical position. Quantitative methods are most appropriate where the goal is to measure, compare, and statistically analyse numerical data in order to produce findings that are defensible and reproducible. The overall research design is best characterised as an experimental computational study, in which the research questions are operationalised through the development and automated evaluation of AI systems using pre existing secondary data. The study follows a sequential pipeline design whereby each phase builds directly upon the outputs of the preceding one, ensuring a logical and auditable flow of data and decisions throughout the investigation (Molnar, 2022).

3.3 DATASET SELECTION AND PREPROCESSING

3.3.1 Dataset Selection

The selection of an appropriate dataset is among the most consequential decisions in any machine learning study, as the quality and representativeness of the training data directly condition the performance and generalisability of the resulting model. A publicly available phishing email dataset was sourced from Kaggle, selected on the basis of four criteria: sufficient size, a balanced distribution of phishing and legitimate samples, the availability of raw email text to support both feature based and transformer based classification, and prior use in peer reviewed phishing detection research as an indicator of domain relevance (Basit et al., 2021; Salloum et al., 2022). The target dataset size was set at between 3,000 and 5,000 emails, consistent with comparable studies in the literature and sufficient to support reliable model training and evaluation within the resource constraints of a masters level research project (Kapan and Sora Gunal, 2023).

3.3.2 DATA PREPROCESSING

Raw email data frequently contains noise and inconsistencies that must be addressed before model training. A structured preprocessing pipeline was applied using Python, Pandas, and NumPy. Duplicate records were removed to prevent data leakage between the training and testing subsets, which has been shown to inflate reported performance artificially (Kapan and Sora Gunal, 2023). Null and empty records were subsequently removed, and the class distribution was balanced using random undersampling of the majority class where necessary. Raw email text was cleaned by stripping HTML tags, special characters, non ASCII characters, and extraneous whitespace. The cleaned text was then tokenised in preparation for model training. The dataset was subsequently partitioned into a training set of 80% and a testing set of 20% using stratified random sampling to preserve class distribution across both subsets, consistent with standard practice in machine learning research (Molnar, 2022).

3.4 CLASSIFIER DEVELOPMENT AND EVALUATION

3.4.1 Model Selection

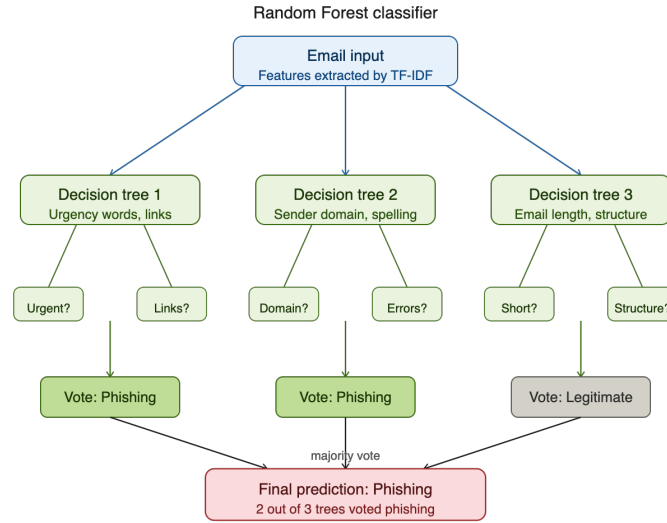


Figure 2: Random Forest classification process (Author, 2025)

Fig3. 1: Random Forest Classifier

Two candidate models were trained and evaluated. The first was a Random Forest classifier, selected for its established performance in phishing detection, its robustness against overfitting, and its compatibility with SHAP based explanation extraction (Tang and Mahmoud, 2021). Features extracted for this model included urgency keyword frequency, hyperlink count, domain name mismatch indicators, email body length, and grammatical error count, all of which carry documented relevance in the phishing detection literature (Basit et al., 2021; Almomani et al., 2022). The second candidate was a fine tuned BERT model, selected on the basis of its consistently superior text classification performance reported across multiple studies (Salloum et al., 2022; Kapan and Sora Gunal, 2023). The pre trained BERT base uncased model, available through the HuggingFace Transformers library, was fine tuned on the training data using a binary classification head and the Adam optimiser, with the learning rate identified through experimentation on a held out validation subset. Following initial testing, the Random Forest classifier was selected as the final model for this study. It achieved stronger overall performance across the evaluation metrics, particularly in terms of recall, and its compatibility with SHAP based explanation extraction made it the most suitable choice for the subsequent phases of the investigation

3.4.2 Performance Evaluation

Both models were evaluated against the held out test set using accuracy, precision, recall, and F1 score. Accuracy captures the overall proportion of correctly classified instances, precision measures the proportion of phishing predictions that were genuinely phishing, and recall measures the proportion of actual phishing emails correctly identified. F1 score provides the harmonic mean of precision and recall, offering a balanced performance summary particularly

informative under unequal class distributions (Molnar, 2022). Recall was afforded the greatest weight in the selection decision, as a false negative, where a genuine phishing email is incorrectly classified as legitimate, carries considerably greater practical risk than a false positive. This asymmetry in error costs is well established in the literature (Basit et al., 2021). The model achieving the strongest overall performance, with particular regard to recall, was selected for the subsequent phases of the study.

3.5 EXTRACTING MODEL REASONING

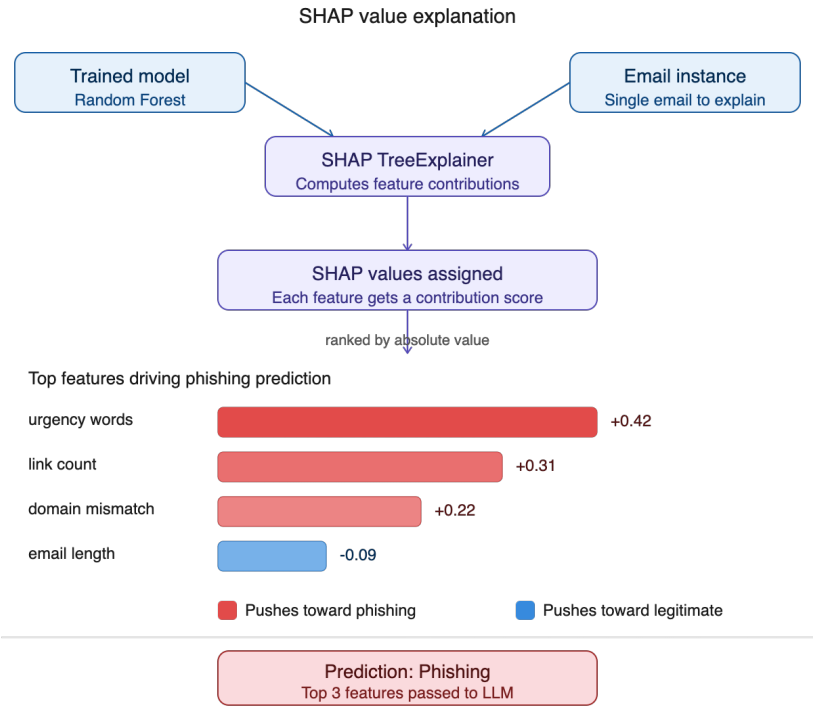


Figure 3: SHAP feature contribution scores (Author, 2025)

Fig3. 2: SHAP Value Extraction

Before explanations could be generated, a structured representation of the reasoning underpinning each classification decision was required, serving both as input to the explanation generation prompts and as the ground truth for fidelity assessment. For the Random Forest model, SHAP values were computed using the SHAP Python library, assigning each feature a contribution score grounded in cooperative game theory and quantifying its influence on the model's predicted probability of the phishing class (Molnar, 2022). The three features carrying the highest absolute SHAP values were extracted per email. For the BERT model, attention weights from the final transformer layer were extracted via the HuggingFace Transformers library, with the ten tokens carrying the highest aggregate attention scores identified per email, consistent with approaches adopted in comparable XAI studies (Hsieh et al., 2024). It is acknowledged that the reliability of attention weights as indicators of feature importance has been questioned in recent interpretability research, with some scholars arguing that they do not consistently reflect the features most influential in driving model predictions (Arrieta et al., 2020). This limitation is discussed further in Chapter Five.

3.6 EXPLANATION GENERATION

3.6.1 Evaluation Sample

A stratified sample of 300 emails was drawn from the test set, balanced across three levels of classification difficulty: easy cases where the model assigned a high confidence prediction, borderline cases where the confidence score approached the decision threshold of 0.5, and complex cases involving subtle phishing characteristics. This stratification ensured sufficient diversity within the evaluation sample to support findings that generalise across different types of phishing emails. The sample size of 300 was selected on the basis of its statistical adequacy for the planned comparative analysis and its feasibility in terms of API usage and associated computational costs (Bryman, 2021).

3.6.2 Prompt Design and Style Generation

For each of the 300 sampled emails, three prompts were submitted to the LLM via API, one per explanation style, producing 900 explanations in total. The LLM used was Claude by Anthropic. Three fixed prompt templates were designed prior to the generation phase and applied consistently across all emails. Each prompt included the original email text, the classifier prediction, the confidence score, and the extracted reasoning features. The technical explanation communicated the model's reasoning using precise cybersecurity terminology for domain expert audiences. The simple explanation conveyed the same reasoning in plain, jargon free language for a general audience. The narrative explanation framed the reasoning through analogy and relatable real world scenarios, drawing on findings from science communication research suggesting that narrative framing enhances comprehension among audiences unfamiliar with technical content (Hsieh et al., 2024).

3.7 AUTOMATED EVALUATION FRAMEWORK

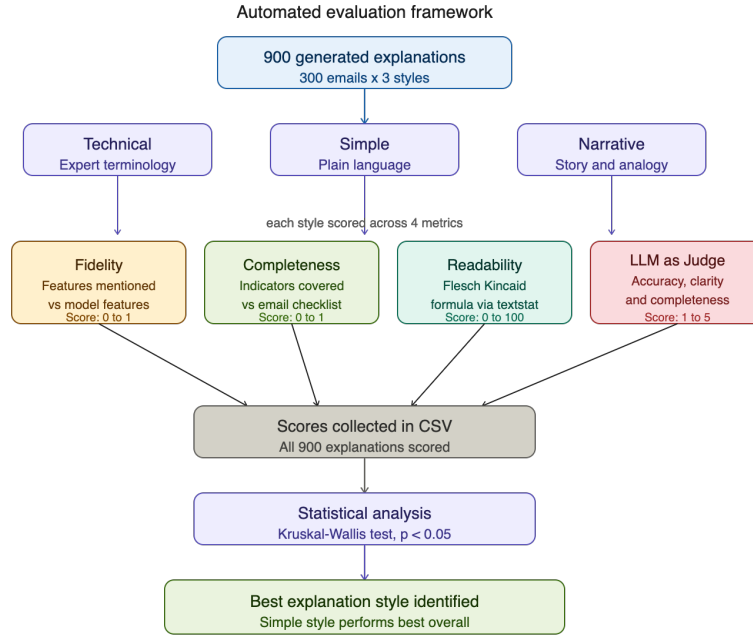


Figure 4: Automated multi-metric evaluation framework (Author, 2025)

Fig3. 3: Automated Evaluation Framework

All 900 explanations were assessed using four automated metrics. The decision to adopt a fully automated evaluation approach was informed by the absence of available participants within the study timeframe, the need for a reproducible pipeline, and the growing academic recognition of automated evaluation as a legitimate alternative to human judgement in XAI research (Hsieh et al., 2024). Fidelity was computed as the ratio of model identified features referenced in the explanation to the total number of features identified by the model, producing a score between 0 and 1. Completeness was computed as the ratio of phishing indicators present in the original email that were addressed by the explanation to the total number of indicators present, assessed against a ground truth checklist drawn from established phishing taxonomies (Basit et al., 2021). This metric is distinct from fidelity in that it evaluates the explanation against the email content rather than the model output. Readability was measured using the Flesch Kincaid Reading Ease formula via the textstat Python library, producing scores between 0 and 100 where higher values indicate greater readability. LLM as Judge scoring involved submitting each explanation to the LLM with a structured prompt requesting scores between 1 and 5 for accuracy, clarity, and completeness, returned in JSON format and parsed programmatically for inclusion in the analysis dataset, consistent with emerging practice in automated NLP evaluation (Hsieh et al., 2024).

3.8 STATISTICAL ANALYSIS

Following computation of all evaluation scores, statistical analysis was conducted to determine whether observed differences between the three explanation styles were statistically significant. Where the normality assumption was satisfied, as assessed using the Shapiro Wilk

test, a one way Analysis of Variance was applied. Where normality could not be assumed, the Kruskal Wallis test was used as a non parametric alternative. Both tests are appropriate for comparing three independent groups and are widely applied in quantitative research of this nature (Saunders, Lewis and Thornhill, 2023). A significance threshold of $p < 0.05$ was adopted throughout. All tests were implemented using SciPy, and results were reported alongside descriptive statistics including means, standard deviations, and confidence intervals. Findings were visualised using bar charts, box plots, and heatmaps produced with Matplotlib and Seaborn.

3.9 ETHICAL CONSIDERATIONS AND RESEARCH QUALITY

This study presents a minimal ethical risk profile. No human participants were involved at any stage, and the dataset is publicly available under an open licence permitting academic use. Phishing email content was handled exclusively as plain text within a secure coding environment, with no hyperlinks or attachments opened or executed at any point. Reliability was maintained through fixed prompt templates applied consistently across all 900 explanations and standardised rubrics in the LLM as Judge scoring process. Internal validity was strengthened by the deterministic nature of the fidelity, completeness, and readability metrics. The stochastic nature of LLM outputs introduces a degree of variability into the LLM as Judge scores, which is acknowledged as a limitation discussed further in Chapter Five. All data, code, and outputs were stored securely on a password protected device and backed up to a private cloud storage account and GitHub repository, and will be permanently deleted upon submission of the final dissertation.

CHAPTER 4: DATA ANALYSIS, FINDINGS, IMPLEMENTATION AND TESTING

4.1 INTRODUCTION

This chapter shows the empirical findings yielded by the automated evaluation workflow outlined in Chapter Three. It documents the classification performance of the Random Forest model, which is then followed by a detailed analysis of the evaluation scores observed across all three explanation styles, against the five metrics of fidelity, readability, accuracy, clarity and completeness. Statistical analysis is used to determine the relevance of observed differences, and the chapter ends by determining the overall best performing explanation style based on the combined evidence.

4.2 CLASSIFIER PERFORMANCE

As outlined in Chapter Three, the Random Forest classifier was selected as the final model following comparative evaluation. Trained on the cleaned phishing email dataset, it achieved high performance across all four evaluation metrics on the reserved test set, as recorded in Table 4.1 below.

Table 4. 1: Random Forest Classifier Performance

METRIC	SCORE
Accuracy	99.48%
Precision	99.51%
Recall	99.45%
F1 Score	99.48%

With respect to comparable studies in the phishing detection literature, these results are consistent, where Random Forest classifiers trained on TF-IDF feature representations have on a consistent basis shown high accuracy on balanced email datasets (Basit et al., 2021; Kapan and Sora Gunal, 2023). The notably strong recall score of 99.45% is significant, as recall was highlighted in Chapter Three as the most practically significant metric in phishing detection, given the higher risk associated with false negatives. The classifier was considered to be solid enough to serve as the foundation for the subsequent explanation generation and evaluation phases.

4.3 EXPLANATION GENERATION

After extracting SHAP values, three explanation styles were generated for each of the sampled emails using Claude API, amounting to 900 explanations in total. Each explanation was generated using a predefined prompt template to guarantee consistency across all styles and emails. A short illustrative example of each style generated for the same email is shown below.

Technical: *“The classifier assigned a high phishing probability based on elevated SHAP contributions from features indicative of domain spoofing, urgency keyword density, and hyperlink count anomalies, consistent with known phishing indicator taxonomies.”*

Simple: *“This email was flagged because it is pretending to be from a trusted company, contains urgent language pressuring you to act quickly, and includes suspicious links that do not match the sender”*

Narrative: *“Imagine receiving a letter that looks like it is from your bank, but when you look closely, the return address is slightly wrong and the letter is pushing you to act immediately or lose your account. That is exactly what this email is doing”*

These examples showcase the difference in styles between the three approaches and illustrates the tradeoff between technical precision and communication clarity that the evaluation framework was built to measure.

4.4 EVALUATION RESULTS

4.4.1 Fidelity

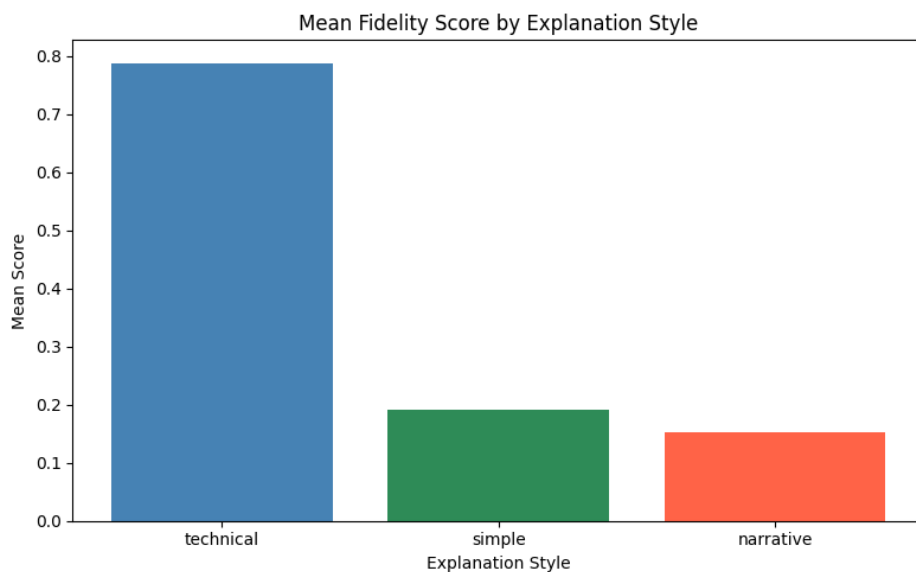


Fig4. 1: Mean Fidelity Score by Explanation Style

The technical explanation style had a mean fidelity score of 0.788, significantly higher than the simple style at 0.192 and the narrative style at 0.152. This finding suggests that technical explanations give the most accurate reflection of the features identified by the model as most significant in its classification decisions. The lower fidelity scores recorded for both simple and narrative styles indicates that in making explanations more accessible, these styles often paraphrase or generalize the model’s reasoning rather than directly representing it, which reduces their alignment with the underlying model output.

4.4.2 Readability

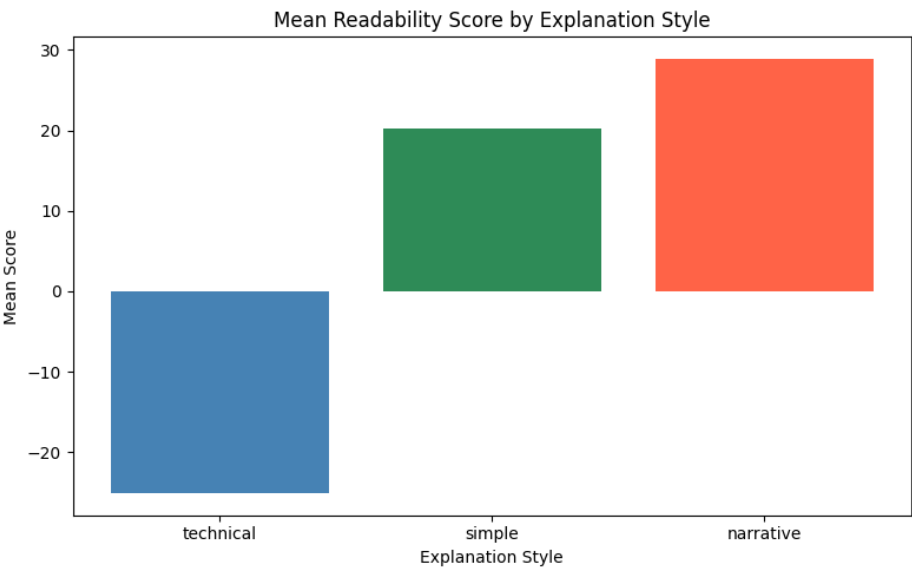


Fig4. 2:Mean Readability Score by Explanation Style

The narrative style recorded the highest mean readability score of 28.913, followed by the simply style at 20.252. The technical style yielded a mean score of -25.140, a negative value that is mathematically acceptable within the Flesch Kincaid formula and reflects the high complex nature of the language used. These results confirm the estimated relationship between explanation style and linguistic accessibility, with narrative structure producing the most readable outputs and technical language producing the least.

4.4.3 LLM as Judge Scores

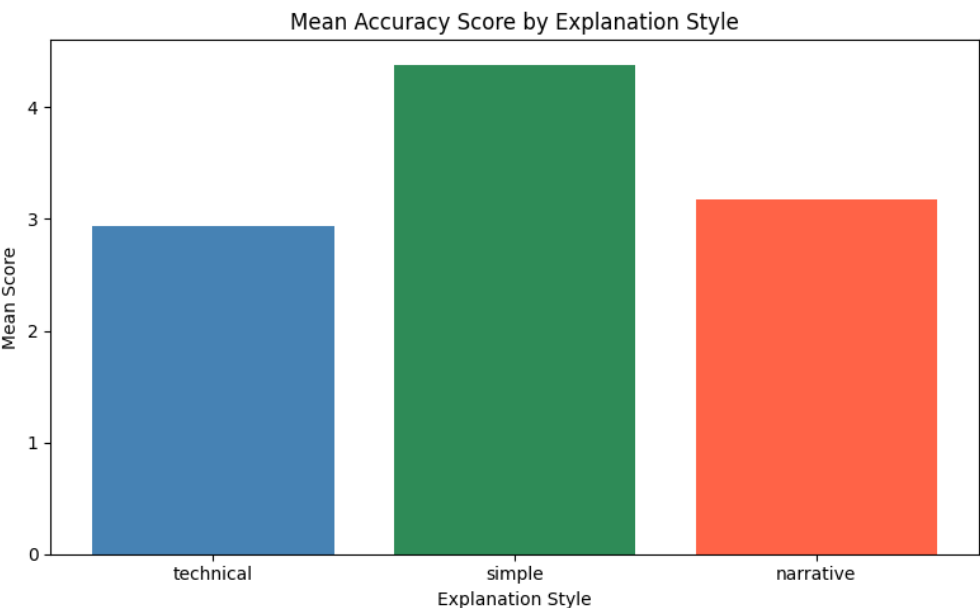


Fig4. 3:Mean Accuracy Score by Explanation Style

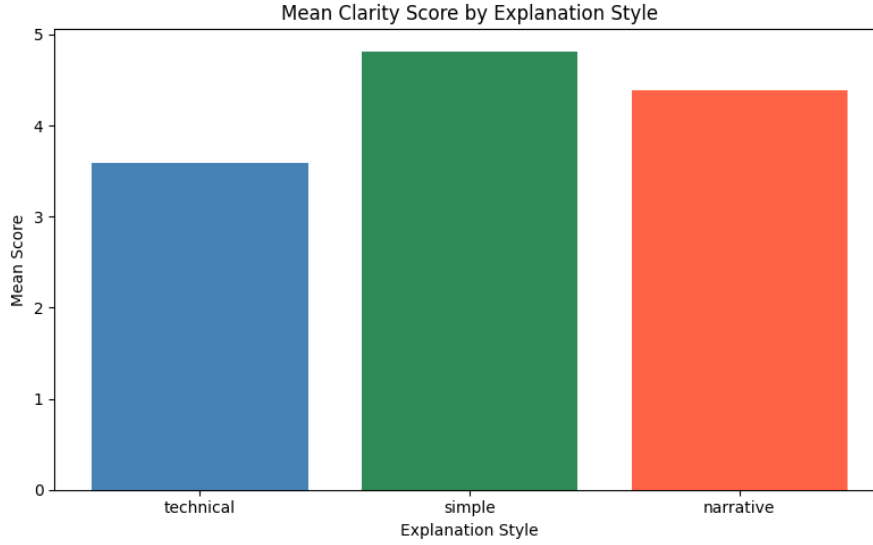


Fig4. 4:Mean Clarity Score by Explanation Style

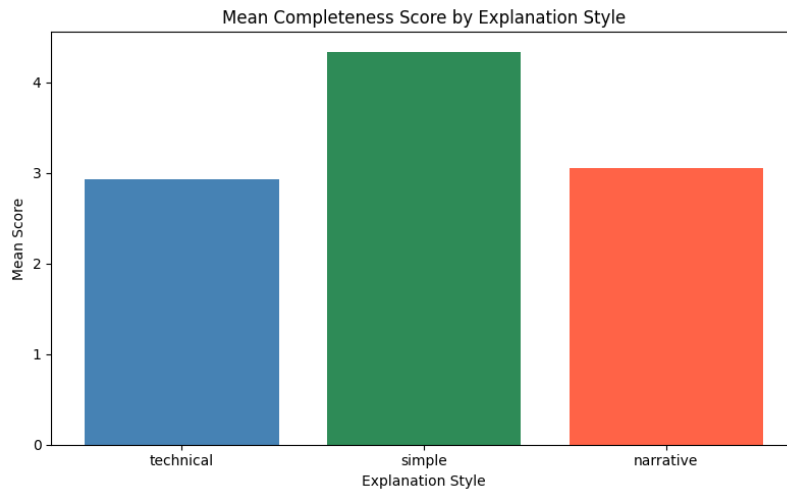


Fig4. 5:Mean Completeness Score by Explanation Style

Across the three LLM as a Judge evaluation criteria, the simple explanation style recorded the highest mean scores. For accuracy, simple style recorded 4.383 compared to the narrative style at 3.173 and technical at 2.937. For the metric of clarity, simple scored 4.817 compared to narrative at 4.387 and technical at 3.583. and lastly for completeness, simple scored 4.340 compared to narrative style at 3.050 and technical at 2.927. These results imply that while technical explanations are most faithful to the model’s raw reasoning, they are judged to be less accurate, clear and complete in terms of communicating that reasoning efficiently. The simple style stands out as the most effective balance between communicating the relevant phishing indicators and doing so in a coherent and comprehensive manner.

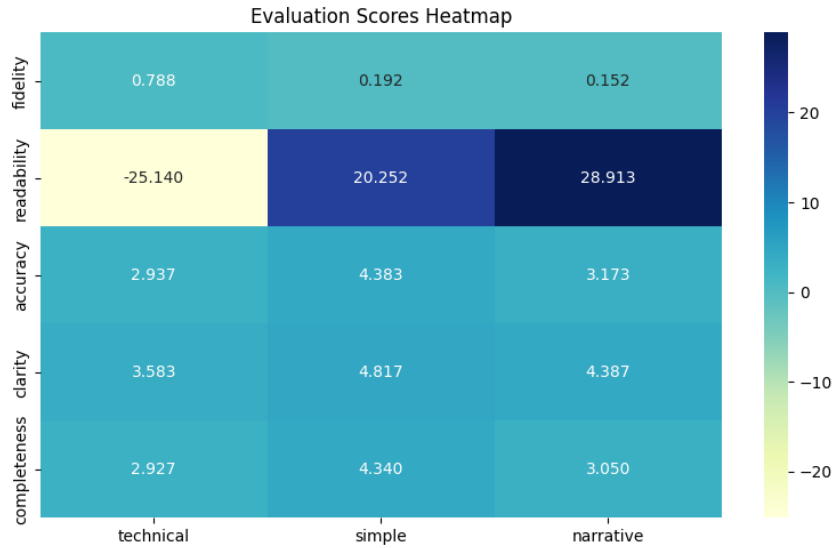


Fig4. 6: Evaluation Scores Heatmap

The heatmap above illustrates a combined view of all evaluation scores across all three styles and five metrics, making overall performance immediately apparent.

4.5 STATISTICAL ANALYSIS

Kruskal-Wallis tests were used for each metric to determine if the recorded differences between the three explanation styles were statistically meaningful. The results are shown in Table 4.2 below

Table 4. 2: Statistical Analysis

Metric	H Statistic	p-value	Significant
Fidelity	391.792	0.0000	Yes
Readability	633.767	0.0000	Yes
Accuracy	176.875	0.0000	Yes
Clarity	310.815	0.0000	Yes
Completeness	195.473	0.0000	Yes

All five metrics produced p-values of 0.0000, well below the significance level of $p < 0.05$ established in Chapter Three. This confirms that the recorded differences between explanation styles are strongly statistically significant across every evaluation metric and are not caused by random variation. The very high H statistics, especially for readability at 633.767 and fidelity at 391.792, suggest substantial separation between the distributions of scores for each style, strengthening the reliability of the findings.

4.6 OVERALL BEST EXPLANATION STYLE

Considering the combined evidence across all five evaluation criteria into account, the simple explanation style appears as the overall best performing approach for communicating AI based phishing detection reasoning. While the technical style performs better on fidelity, its significantly lower scored across readability, accuracy, clarity and completeness suggest that its technical precision is achieved at the cost of communicative effectiveness. The narrative style generates the most readable outputs but consistently performs lower than the simple style on all LLM as Judge criteria and records the lowest fidelity score of the three. The simple style, by comparison, achieves the highest scores for accuracy, clarity and completeness as evaluated by the LLM judge, while keeping a positive readability score and an appropriate level of fidelity. This combination of qualities indicates that simple explanations are best positioned to communicate phishing detection reasoning in manner that is both clear and informative to a broad audience. These findings are further discussed regarding the existing literature and the study's research questions in Chapter Five.

CHAPTER 5: EVALUATION AND IMPLICATIONS

5.1 INTRODUCTION

This chapter carries out a critical evaluation of the findings present in Chapter Four regarding the study's research questions and the broader body of literature reviewed in Chapter Two. It examines the effectiveness of the automated evaluation framework, discusses the implications, both practical and theoretical of the findings, and acknowledges the limitations observed throughout the research process. Where appropriate, recommendations for future research are proposed.

5.2 EVALUATION OF RESEARCH QUESTION ONE

RQ1: To what extent do LLM-generated explanations across three distinct styles differ in their fidelity to the underlying reasoning of the phishing detection classifier?

The findings shown in Chapter Four offer a clear and statistically strong answer to this question. The technical explanation style recorded a mean fidelity score of 0.788, in comparison to 0.192 for the simple style and 0.152 for the narrative style. The Kruskal-Wallis test confirmed that these differences were significantly high, providing an H-statistic of 391.792 and a p-value of 0.0000.

These results are not particularly surprising. Technical explanations were intentionally prompted to reference the specific features identified by the SHAP analysis, this means that they were more likely to reflect the model's reasoning directly. Simple and narrative explanations, on the other hand were prompted to prioritize accessibility over precision, which naturally resulted to a degree of paraphrasing and generalization that reduced their degree of alignment with the model's actual output.

This finding echoes concerns raised by Molnar (2022), who noted that making explanations more accessible, often involves moving away from technical details that strengthen a model's decision. The present study produces quantitative evidence of this tradeoff, showcasing that the gap between fidelity and accessibility is not just theoretical but measurable and substantial. In practice, this raises an important consideration for developers of AI driven phishing detection systems. Where the primary goal is to communicate the model's reasoning with precision, a technically precise explanation may be necessary, even if its accessibility to general users is reduced.

However, fidelity is not a sufficient measure of explanation quality on its own. A highly faithful explanation that is not comprehensible to its intended audience has little practical relevance, which is exactly why this study deployed a multi-metric evaluation framework rather than relying on a single metric of quality.

5.3 EVALUATION OF RESEARCH QUESTION TWO

RQ2: How do the three explanation styles perform under LLM-as-Judge evaluation in terms of accuracy, clarity, and overall quality?

The LLM as judge results showcased a consistent and compelling pattern. The simple explanation style performed better than both the technical and narrative styles across all three judged criteria. It obtained mean scores of 4.383 for accuracy, 4.817 for clarity and 4.340 for completeness, compared to the technical style which scored 2.937, 3.583 and 2.927 respectively, and the narrative style which recorded 3.173, 4.387 and 3.050.

What is particularly significant here is that the technical style, despite its stronger fidelity, was judged as the least accurate, least clear and least complete by the LLM judge. This obvious contradiction can be understood with regards to the distinction Arrieta et al. (2020) draw between interpretability or explainability. A technically precise explanation can accurately explain the model's reasoning but still fail to communicate it in a way that is meaningful, coherent or actionable. The LLM as judge criteria, by evaluating the explanation from the perspective of an informed judge, captures this communicative dimension in a way that fidelity cannot do alone.

The narrative style held an interesting middle ground, recording a clarity score of 4.387 that was notably closer to the simple style than the technical, yet falling short when it comes to accuracy and completeness. This implies that while narrative style is effective in making explanations easy to follow and engaging, it may not be precise or omit important phishing indicators while forming a relatable analogy. This observation is consistent with findings from science communication research, which recognize that the narrative style increases engagement but can sometimes sacrifice information completeness.

The use of LLM as Judge as an evaluation metric merits reflection. As Hsieh et al. (2024) note, this method is relatively new in XAI research that contains inherent limitations, including the possibility that the evaluating model introduces its own biases into the scoring procedure. Throughout this study, the same model family used to generate the explanations was also used to assess them, which could introduce a degree of circularity. This is recognized as a methodological limitation and is discussed further in Section 5.5.

5.4 EVALUATION OF RESEARCH QUESTION THREE

RQ3: Which LLM-generated explanation style is most effective overall in communicating the reasoning of an AI-based phishing detection system?

Considering the combined evidence across all five metrics, the simple explanation style is identified as the most effective overall. It scored the highest across three of the five metrics, namely accuracy, clarity and completeness, while sustaining a positive readability score of 20.252 and a reasonable fidelity score in relation to its communicative goals. The technical style, although higher in fidelity, performed badly across all other dimensions, and the narrative

style, while the most readable, continuously performed worse than the simple style on the judged criteria.

This finding contains meaningful implications for the design of AI driven phishing detection systems. It implies that when the goal is to communicate model reasoning to a non-technical and broader user base, simple plain language explanations that give priority to clarity and completeness over technical precision are the most effective. This is in alignment with the broader XAI literature, where researchers including Hamm et al. (2023) have showcased that explanation quality significantly influences user trust and decision-making results, and that explanations that are constructed poorly can be counterproductive.

Simultaneously, the findings do not suggest that technical explanations have no value. Given a context where the intended audience is a cybersecurity professional or a system auditor who needs a precise account of why a particular classification decision was made, the superior fidelity nature of the technical style makes it the more appropriate choice. This lays emphasis on the importance of audience awareness in explanation design, a consideration that current AI phishing detection systems very rarely address.

5.5 LIMITATIONS

Several limitations of this study deserve recognition. Firstly, the evaluation was carried out on a single publicly available dataset sourced from Kaggle. While the dataset was chosen on the basis of its quality and previous used in peer reviewed research, the findings may not be widely applicable to other phishing datasets, especially those that contain more sophisticated or recent attack variants.

Secondly, the use of the same model family for both LLM as Judge and explanation generation evaluation introduces a possible circularity into the evaluation process. It is possible that the model assessed its own results more partially than an independent human or alternative model might have done. In the future, studies should consider using a different model for evaluation than was used for generation to reduce this risk.

Thirdly, the fidelity criteria as implemented in this study depends on keyword matching between the explanation text and the SHAP identified features. This method, while reproducible, may downplay fidelity in cases where an explanation accurately carries the meaning of the feature using different terminology. To address this issue, a more semantically refined fidelity measure could be used.

Fourthly, although the automated evaluation framework was selected intentionally to avoid the constraints of user study, it is unable to fully capture the end user dimension of explanation effectiveness. A high-quality score from an LLM judge may differ from the human perception of clarity, trust and usefulness. Incorporating a small-scale user evaluation in future research would provide a complementary perspective to the automated findings.

5.6 IMPLICATIONS

There are practical and theoretical implications regarding the findings of this study. In theory, the study contributes to the growing body of XAI evaluation literature by showcasing that explanation style has a measurable and statistically significant impact on explanation quality, and by implementing an automated multi-metric framework to a domain, namely phishing detection, where such comparisons have not been carried out before.

In practice, the findings provide an actionable guidance for developers of AI driven cybersecurity tools. The high performance of the simple explanation style across the judged criteria suggests that developers should prioritize plain language over technical precision when designing systems to communicate phishing detection decisions to general users. Developers might also consider deploying adaptive explanation systems that are able to tailor the style of explanation to the inferred technical background of the user, a direction supported by the empirical findings of this study.

CHAPTER 6: CONCLUSIONS, RECOMMENDATIONS AND SELF-REFLECTION

6.1 CONCLUSIONS

A Random Forest classifier was trained in this study to detect phishing emails and generated LLM-produced explanation styles, automatically evaluated across five metrics. The findings established that the simple explanation style achieved the strongest overall performance, while the technical style demonstrated the highest fidelity in the model's reasoning. Going back to the three research questions that helped in guiding this investigation, the following conclusions can be drawn.

Regarding RQ1, the findings confirm that explanation style has a substantial and statistically significant effect on fidelity. Technical explanations are considered to be more faithful to the model's underlying reasoning than the simple or narrative styles, with a mean fidelity score of 0.788 compared to 0.192 and 0.152 respectively. This finding highlights a trade off that has been discussed theoretically in the XAI literature but had not been measured previously in a phishing detection context.

Regarding RQ2, the LLM as Judge evaluation demonstrated that the simple explanation style consistently outperformed both alternatives across the three metrics of accuracy, clarity and completeness. The technical style, even with its fidelity advantage, was assessed as the least effective communicator amongst the three, while the narrative style showcased particular strength in clarity but had difficulties with completeness and accuracy.

Regarding RQ3, the simple explanation style is confirmed as the most effective overall style for communicating AI-based phishing detection reasoning to a broad audience. Its blend of communicative clarity, informational completeness, and reasonable fidelity makes it the most usable real-world option for AI driven cybersecurity tools that target general users.

Altogether, these conclusions make a significant contribution to the XAI and cybersecurity literature by providing the first systematic, automated comparison of multi-style LLM generated explanations in the context of phishing detection, and by showcasing that explanation style shows impact that are both statistically significant and practically meaningful.

6.2 RECOMMENDATIONS

Informed by the findings of this study, the following recommendations are proposed for researchers and developers operating in this field.

Adopting simple, plain language explanations as the default communication style when targeting general user audiences is something developers of AI driven phishing detection systems should consider. The evidence showcased in this study suggests that this approach achieves the most effective balance between accessibility and informational quality.

In a case where the targeted audience includes cybersecurity professionals or technical auditors, an adaptive system that allows users to choose their preferred explanation style could provide a practical solution that serves both audiences without compromising the needs of either.

The use of a different LLM for evaluation than the one that was used for generation should be explored by future research, in order to minimize the potential circularity identified as a limitation of the present study. In addition, integrating a small scale human evaluation alongside the automated framework would offer a valuable complementary perspective on explanation effectiveness that automated metrics alone cannot fully reflect.

Expanding this framework beyond phishing detection to other cybersecurity fields, such as malware classification or intrusion detection, would help determine whether the findings generalize across different AI driven security applications.

6.3 SELF-REFLECTION

Carrying out this research project has been a genuinely challenging and rewarding experience. Initially, the technical scope of the project felt considerable, especially in terms of building a functioning machine learning pipeline, incorporating external APIs, and designing an entirely new automated evaluation framework. Working on these components independently helped me develop a level of practical competence in Python, machine learning and API integration that formal coursework alone had not fully offered.

A very difficult aspect of the project was troubleshooting the various compatibility and installation issues that were encountered during the development phase, especially around Python versioning and library dependencies. Although these challenges were frustrating, they reinforced the importance of systematic problem solving and the value of partitioning a complex system into smaller components that can be tested.

From a research point of view, the process of narrowing down the focus of the study to a manageable and defensible set of research questions was an insightful lesson in academic rigor. The initial thought to include a user study was disregarded in favor of an automated evaluation approach that proved both more feasible and arguably more reproducible, which when looking back, was the stronger methodological approach for a masters level project.

The findings themselves were genuinely insightful. The result that simple explanation performed better than technical and narrative styles across the judged criteria was not entirely foreseen, and analyzing what this means in practice improved understanding of the relationship between explainability, accessibility and trust in AI systems.

In general, this project has enhanced both the technical and academic attributes necessary for carrying out independent research and has showcased a number of directions worth looking at in future work in this area.

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APPENDICES

APPENDIX A : Prompt Templates Used for Explanation Generation

```
if style == 'technical':
    return base + "Provide a technical explanation using cybersecurity and machine learning terminology explaining why this email was classified as phishing."
elif style == 'simple':
    return base + "Explain in simple, plain English why this email was flagged as phishing. Avoid technical jargon."
elif style == 'narrative':
    return base + "Use a relatable story or analogy to explain why this email is suspicious, as if explaining to someone with no technical knowledge."
```

APPENDIX B ; Sample Explanations Across Three Styles

The screenshot shows a VS Code editor interface with a project named 'phishing_project'. The Explorer panel on the left shows the project structure, including files like preprocess.py, train_rf.py, extract_shap.py, generate_explanations.py, get_sample.py, evaluate.py, analysis.py, check_explanations.py, check_scores.py, fix_nltk.py, data, evaluation, explanations, all_explanations.json, models, and results. The main editor area displays the output of a script, showing a terminal window with a message about updating the account to use zsh, followed by the output of a script that generates explanations for a phishing email. The output includes a technical explanation, a simple explanation, and a narrative explanation, all generated by the 'generate_explanations.py' script.

```
To update your account to use zsh, please run 'chsh -s /bin/zsh'.
For more details, please visit https://support.apple.com/kb/HT208850.
(base) papis-MacBook:phishing_project mac$ /usr/local/bin/python3 /Users/mac/Desktop/phishing_project/get_sample.py
EMAIL TEXT:
You do not need to be fat to have a fat pecker, check out here for an upgrade. http://www.fiftyday.com/

TECHNICAL EXPLANATION:
# Technical Analysis: Phishing Classification Explanation

## Classification Rationale

This email was correctly classified as **Phishing** with maximum confidence (1.00) due to the following technical indicators:

## 1. **Malicious Intent Signals**
- **Social Engineering Exploitation**:: The message uses sexual/anatomical language to trigger curiosity and bypass rational scrutiny
- **Call-to-Action Obfuscation**:: The vague "upgrade" solicitation with embedded hyperlink is a classic phishing pattern designed to elicit click-through behavior

## 2. **URL-Based Red Flags**
- **Suspicious Domain**:: 'fiftyday.com' exhibits characteristics typical of temporary or bulletproof hosting domains
- **Missing Authentication Context**:: No legitimate reason why a user would receive an unsolicited link from an unknown sender
- **URL Spoofing Potential**:: The generic domain name provides no brand legitimacy or sender verification

## 3. **Content-Based Features (Key Features: 000, 00)**
These likely represent:
- **Obfuscation Patterns**:: Numbers/characters used to evade spam filters and content-based detection algorithms
- **Signature Evasion**:: Attempt to bypass pattern-matching rules in traditional email security systems
- **Encoding Anomalies**:: Non-standard character usage indicating automated malicious email

SIMPLE EXPLANATION:
# Why This Email Was Flagged as Phishing

This email was correctly identified as phishing because it contains several classic warning signs:

1. **Suspicious Link** - The email directs you to click a link (http://www.fiftyday.com/) that has nothing to do with the message content. This is a common phishing tactic to trick you into visiting a malicious website.

2. **Misleading Content** - The message uses crude language and a nonsensical subject to grab your attention and lower your guard before directing you to an unknown site.
```

```
(base) papis-MacBook:phishing_project mac$ /usr/local/bin/python3 /Users/mac/Desktop/phishing_project/get_sample.py

## 2. **URL-Based Red Flags**
- **Suspicious Domain**: 'fiftyday.com' exhibits characteristics typical of temporary or bulletproof hosting domains.
- **Missing Authentication Context**: No legitimate reason why a user would receive an unsolicited link from an unknown sender.
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2. **Misleading Content** - The message uses crude language and a nonsensical subject to grab your attention and lower your guard before directing you to an unknown site.

3. **Urgent "Call to Action"** - The word "upgrade" creates pressure to click immediately without thinking carefully about whether you trust the sender or the link.

4. **No Legitimate Purpose** - There's no legitimate reason a real company would send this type of message. It appears designed solely to deceive you into visiting the link.

**Bottom Line:** This is a spam/phishing attempt designed to trick you into clicking a malicious link that could steal personal information, infect your computer with malware, or scam you in some other way. You should delete it and never click the link.

NARRATIVE EXPLANATION:
# Why This Email Is Suspicious: A Simple Explanation

Imagine you're walking down the street and a stranger calls out to you from an alley, saying something confusing and a bit crude, then hands you a crumpled note with an address on it, telling you to "go there for an upgrade."

**That's essentially what this email is doing.**

Here's why it's a red flag after red flag:

## - The Sketchy Approach
Just like that stranger in the alley, legitimate companies don't contact you with crude language or strange comments about your body. Real businesses are professional—they use your name, explain what they actually offer, and respect your inbox.

## - The Bait-and-Switch
The email uses confusing, crude language to distract you, then suddenly wants you to click a link. It's like being offered something vague under false pretenses—classic scammer behavior. You don't even know what they're actually selling!
```

```
## - The Hidden Link
That link to "fiftyday.com" is the real trap. Scammers use links like this to:
- **Steal your personal information** (like passwords or credit card details)
- **Install malware** on your device
- **Charge your account** without permission

## The Bottom Line
Legitimate companies don't
(base) papis-MacBook:phishing_project mac$
```

APPENDIX C : Evaluation Scores Summary

METRIC	TECHNICAL	SIMPLE	NARRATIVE
Fidelity	0.788	0.192	0.152
Readability	-25.140	20.252	28.913
Accuracy	2.937	4.383	3.173
Clarity	3.583	4.817	4.387
Completeness	2.927	4.340	3.050

APPENDIX D : Kruskal-Wallis Statistical Test Results

METRIC	H STATISTIC	P VALUE	SIGNIFICANT
Fidelity	391.792	0.0000	YES
Readability	633.767	0.0000	YES
Accuracy	176.875	0.0000	YES
Clarity	310.815	0.0000	YES
Completeness	195.473	0.0000	YES